DAVINA WHATA

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PROFESSIONAL SUMMARY

GIS professional and M.S. Geographic Information Science candidate specializing in utility infrastructure mapping, spatial data integration, and geospatial quality assurance. Brings cross-industry experience in telecommunications, municipal infrastructure, and environmental consulting, with expertise in transforming complex spatial datasets into accurate, engineering-ready GIS deliverables.

TECHNICAL SKILLS

GIS Platforms: ArcGIS Pro, ArcGIS Online, ArcMap, QGIS, VETRO FiberMap, Fiber Manager, Field Maps, Survey123

Spatial Data Engineering: Geodatabases, GIS data integration, utility records, feature/attribute editing, topology QA/QC, network connectivity, raster/vector processing, spatial statistics

Python & Data Tools: Python, ArcPy, GeoPandas, pandas, GDAL, Jupyter Notebook, GPS/GNSS, CAD-to-GIS, KML/KMZ, remote sensing

PROFESSIONAL EXPERIENCE

GIS Technician Intern | Thayer Power & Communication **May–Aug 2026** • Updated and validated fiberoptic network designs across 8 projects in 4 Indiana counties using ArcGIS Pro and VETRO FiberMap.

- Resolved fiber circuitry and connectivity errors by tracing customer drops through splitters and splice cases to feeder tie points.
- Corrected topology, slack-loop, handhole, and cable-routing errors, preparing network designs for engineering review.

Graduate Research Assistant | Michigan Technological University **Aug–Dec 2025**

- Validated 1,000+ GPS/GNSS-referenced wildlife records, correcting coordinate and attribute inconsistencies for ecological research.
- Managed geospatial QA/QC for a 4-person research team, maintaining organized records and producing GIS maps.

GIS & Environmental Assessment Support | LW Consultants **Dec 2024–Jul 2025**

- Delivered 5 GIS mapping products across 5 environmental projects supporting Environmental Impact Assessments and regulatory submissions.
- Integrated environmental and land-use datasets, correcting spatial and attribute errors before regulatory review.

GIS Support Assistant | Barkly West Municipality **May 2023–Aug 2024**

- Updated municipal water, sewer, stormwater, and road asset layers using GPS/GNSS observations and engineering records.
- Corrected 100+ CAD-converted utility records, resolving topology and attribute inconsistencies to improve GIS data quality.

SELECTED GEOSPATIAL PROJECTS Utility Data Integration & Validation — Network and municipal assets

Integrated engineering and field records into GIS features; checked attributes, topology, and network consistency across utility datasets.

Spatial Method Evaluation — Geostatistics and remote sensing

Compared IDW, Natural Neighbor, and kriging using cross-validation; analyzed Landsat imagery and spatial patterns with Moran's I and kernel density.

Python Geospatial Processing — ArcPy, GeoPandas, and GDAL

Used Python GIS libraries for data extraction, proximity analysis, raster resampling, and processing workflows; inspected outputs for accuracy.

LEADERSHIP & ACADEMIC EXPERIENCE Academic Tutor & Mentor | Sol Plaatje University Feb 2023–Nov 2024

- Tutored undergraduate students in GIS Techniques and Environmental Archaeology, clarifying course concepts and supporting learning.
- Mentored students through academic guidance, individualized support, and collaborative learning. **EDUCATION**

Michigan Technological University | M.S. Geographic Information Science | Expected Dec 2026 Sol Plaatje University | Environmental Science | GPA: 3.83 | May 2025